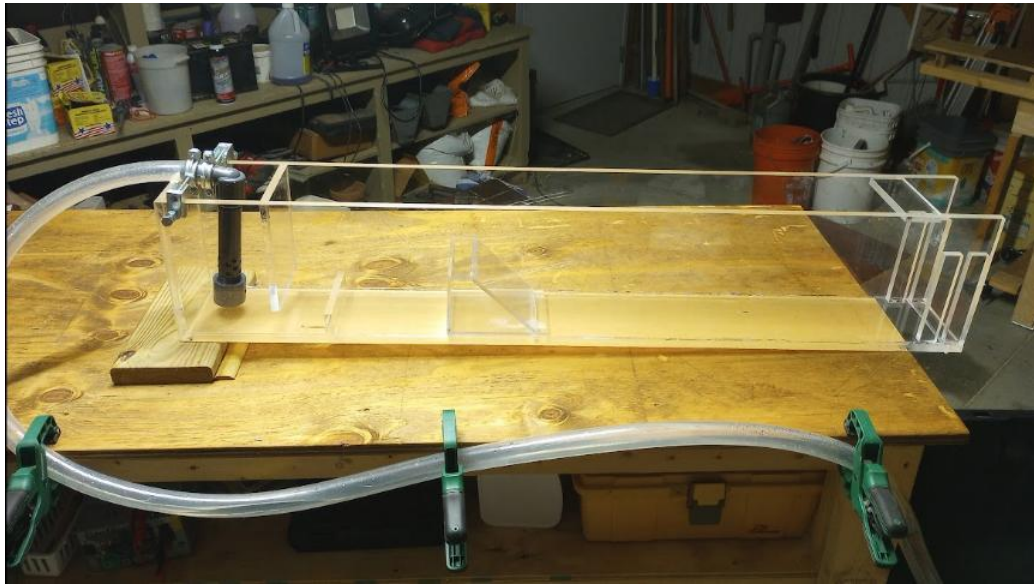


Instructions for Constructing a Low-Head Dam Demonstration Flume

A hydraulic flume is an excellent tool for demonstrating how low-head dams create dangerous recirculating hydraulics (“hydraulic rollers”) and why people should avoid these structures. The flume described below is small, portable, and suitable for classrooms, water festivals, fairs, and other educational events. It also helps spark interest in dam engineering and hydraulics.



The overall flume dimensions are 6" x 9" x 46.5".

This flume was made from ½" thick acrylic. The ½-inch acrylic provides sufficiently rigid sidewalls to maintain a consistent width and allows removable hydraulic elements to fit securely between the sidewalls. If the flume will use only a permanently installed weir, 3/8-inch acrylic may be sufficient. The low-head dam weir insert for this demonstration was made from ¼" thick acrylic.

The flume is designed to be portable and simple enough to construct with commonly available materials and tools. Minor modifications may be necessary depending on the materials, fittings, pump, and construction methods used.

This flume was inspired by the flume in Practical Engineering's YouTube video "The Most Dangerous Dams".

https://www.youtube.com/watch?v=GVDpgphHhAE&list=PLTZM4MrZKfW_0exE8eEXgjYahGlkQ_NUa&index=8

Also see, the Fluids Explained YouTube video "I Built a Lab Water Flume in My Office at Home"

<https://www.youtube.com/watch?v=sppaBqpIT-w>

General Assembly Sequence

1. Cut acrylic pieces
2. Deburr/soften cut edges
3. Assemble and weld the sidewalls to the bottom
4. Weld endwall to sidewalls
5. Install the internal baffles
6. Install the stoplog guides
7. Construct the removable weir
8. Allow solvent welds to cure for 24-hours before testing
9. Test the flume for leaks
10. Install the pump and hose system
11. Install the removable weir

Cutting the Acrylic

Acrylic Cut List for Flume

- 8-1/2" x 46-1/2" x 1/2" (2) Sidewalls
- 8-1/2" x 5" x 1/2" (1) End wall
- 5-31/32" x 46-1/2" x 1/2" (1) Bottom*
- 5" x 7" x 1/2" (1) Upper Baffle
- 5" x 2" x 1/2" (2) Lower Baffle & Top Strut
- 6" x 1" x 1/2" (4) Stoplog Guides
- 4-15/16" x 1-3/8" x 1/2" (3) Removable Stoplogs

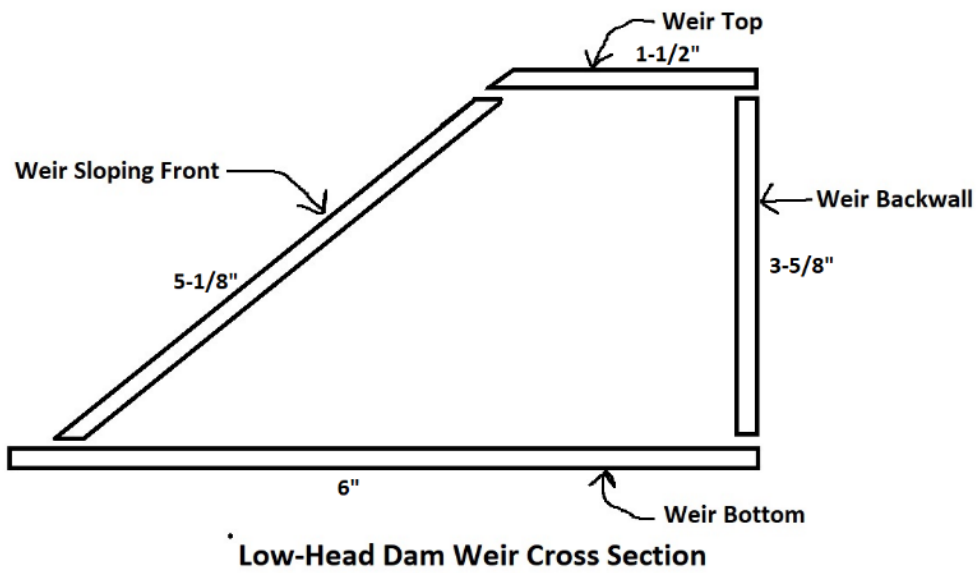
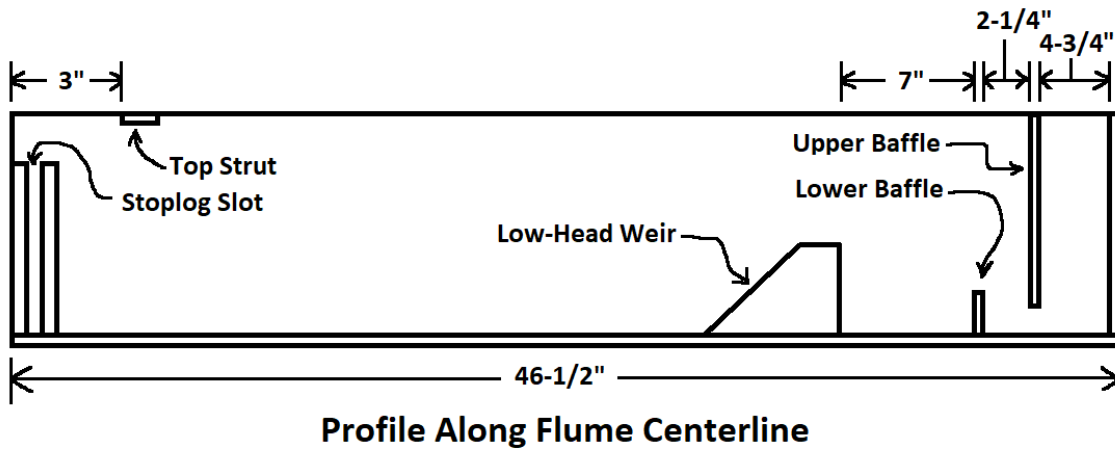
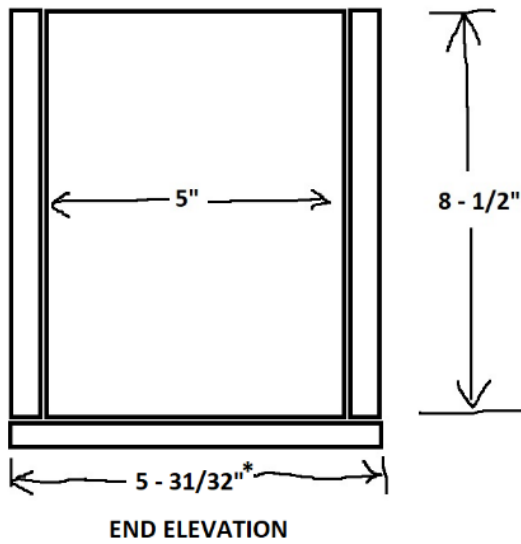
*Assumes "1/2-inch" acrylic is actually 0.485" thick. Adjust width if using a different thickness.
(e.g. $5" + 2 \times 0.485" = 5.97"$ or 5-31/32")

Acrylic Cut List for Removable Weir

- 4-15/16" x 6" x 1/4" (1) Weir Bottom
- 4-15/16" x 1-1/2" x 1/4" with 45-degree bevel on one 5" side (1) Weir Top
- 4-15/16" x 3-5/8" x 1/4" (1) Weir Back Wall
- 4-15/16" x 5-1/8" x 1/4" with parallel 45-degree bevels on each 5" side (1) Weir Sloping Front

Cutting a 1/2-inch acrylic sheet can be difficult. There are many YouTube videos on the topic, especially for people who are constructing fish aquariums. You may also be able to find a local business or plastic supplier that is willing to cut the acrylic for you.

The cut edges of acrylic can be sharp. You can use a metal straight edge, such as a ruler, to drag along the corners and soften the edges.



Assembling and Welding

After all the pieces are cut, it is time to weld the pieces together. Find a well-ventilated area for assembly.

Here is what you will need:

- Weld On 3
- Weld On 16 (for filling wide gaps)
- Applicator – suggest using syringe with a needle.
- Tape that won't react with Weld On
- Clamps
- Small speed square (it helps to have 2)
- Safety glasses
- Nitrile gloves

There are many YouTube videos demonstrating the welding process. Here are two that may be helpful:

1. "HOW TO: Build An Aquarium 4/6"
<https://www.youtube.com/watch?v=3rsUbzVwaUo&list=PLcWeEm9w62mYYIBtX2gd98mXp67fKoiYb&index=4>
2. "HOW TO BUILD AN AQUARIUM (ACRYLIC) fish tank!"
https://www.youtube.com/watch?v=rho_cQXD7yA

See photos at the end of these instructions showing the welding process.

Testing for Leaks

After construction and allowing adequate time for the solvent welds to cure, slowly fill the flume with water and inspect all seams for leaks. Allow any identified leaks to dry completely before making repairs. Consider using a hair dryer to ensure the seam is dry. Small seeps can usually be repaired by tipping the flume back at a 45-degree angle and resting it on a block of wood. Then, run a small amount of solvent along the seam, allowing the solvent to penetrate the seam and find the leak.

Pump and Hose Setup for Water Delivery

Water will be pumped from the storage container through the flexible hose and into the upstream end of the flume. Water exiting the flume drains back into the storage container, creating a recirculating system.

Items needed:

1. Pump - such as VIVOHOME 100W Pond Pump, 700-1800GPH Adjustable Flow Submersible Water Pump
2. Large container like a trash can (40 gallon) or tote (25 gallon) to catch and hold water. The pump will go in the bottom of the trash can or tote.
3. Hose – 10 feet of 1-inch I.D. x 1-1/4" O.D. flexible clear vinyl tubing
4. 2 or 3 adjustable clamps to hold the hose in place as it runs along the back side of a table.
5. 3/8" top beam clamp (2) for clamping the pipe supports to the side of the flume
6. 3-inch long, 3/8-inch all thread rod (2) for attaching pipe supports to beam clamp
7. 1-inch galvanized split ring tubing hanger or pipe support (2) to mount the hose to the flume
8. 1-inch PVC Insert Fitting, Threaded Female to Barbed (for connecting hose to pump)



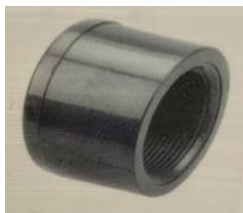
9. The following parts were used to build the water diffuser at the outlet end of the hose.
 - a. 1-inch PVC Insert Fitting, Threaded Female Elbow to Barbed.



- b. 6-inch x 1-inch PVC Riser with Threaded Male Ends



- c. 1-inch PVC Threaded Cap



10. 7/16" Drill Bit

See photos at the end of these instructions for the pump and hose setup.

Installing the Weir

The weir can either be welded in place to create a permanent installation or designed to be removable. If the weir is removable, several options can be used to hold it securely in place, including a friction fit, adding weight to hold it down, temporarily securing it with a small amount of silicone, or incorporating slots into the flume design so the weir can slide into place.

Additional Notes:

1. An adjustable-flow pump is recommended so the flow rate can be adjusted as needed to achieve the desired hydraulic conditions. Flow can also be adjusted by adding a ball valve to the hose. When selecting a pump, keep in mind that the actual flow rate will be less than the pump's rated capacity due to head losses from fittings, pump elevation, and hose length. A minimum flow rate of 10 gpm at the flume is recommended.
2. Dimensions shown are based on the prototype flume. Some dimensions, particularly those associated with the removable hydraulic elements, are important for proper fit and function. Minor adjustments may be necessary to account for the actual acrylic thickness and fabrication tolerances. Plan to make minor adjustments by trimming, cutting, or sanding the acrylic as needed to achieve a proper fit. A miter saw with a good carbide blade will often work for small cuts. It is recommended to reinforce the acrylic with same masking tape before you make the cut.
3. A hard-sided golf travel case can be used to transport the flume.
4. For questions you can contact:
Tim Gokie, P.E.
tim.gokie@nebraska.gov
5. If you build a flume, be sure to send Tim a picture.

These instructions are provided as a general guide and are not intended to address every aspect of the construction, transportation, setup, or operation of the flume. Please use appropriate safety measures and good judgment at all times. Follow all manufacturers' instructions. Users are responsible for determining and following appropriate safety practices for their particular circumstances and assume all risks associated with the construction and use of the flume. The authors and organizations providing these instructions assume no responsibility or liability for any injury, damage, or loss resulting from the construction, transportation, setup, or use of the flume.

Photos of Flume Construction



Photo 1: Cut $\frac{1}{2}$ " acrylic for flume construction ready for assembly.



Photo 2: Ready to weld flume walls to flume bottom. 5-inch spacers were used with clamps to hold walls in place.



Photo 3: Ready to weld flume walls to flume bottom. 5-inch spacers used with clamps to hold walls in place.



Photo 4: Weld On solvent and bottle applicator. (Consider using syringe with needle instead.)



Photo 5: Using bottle applicator to place solvent inside seam to weld sidewalls to bottom.

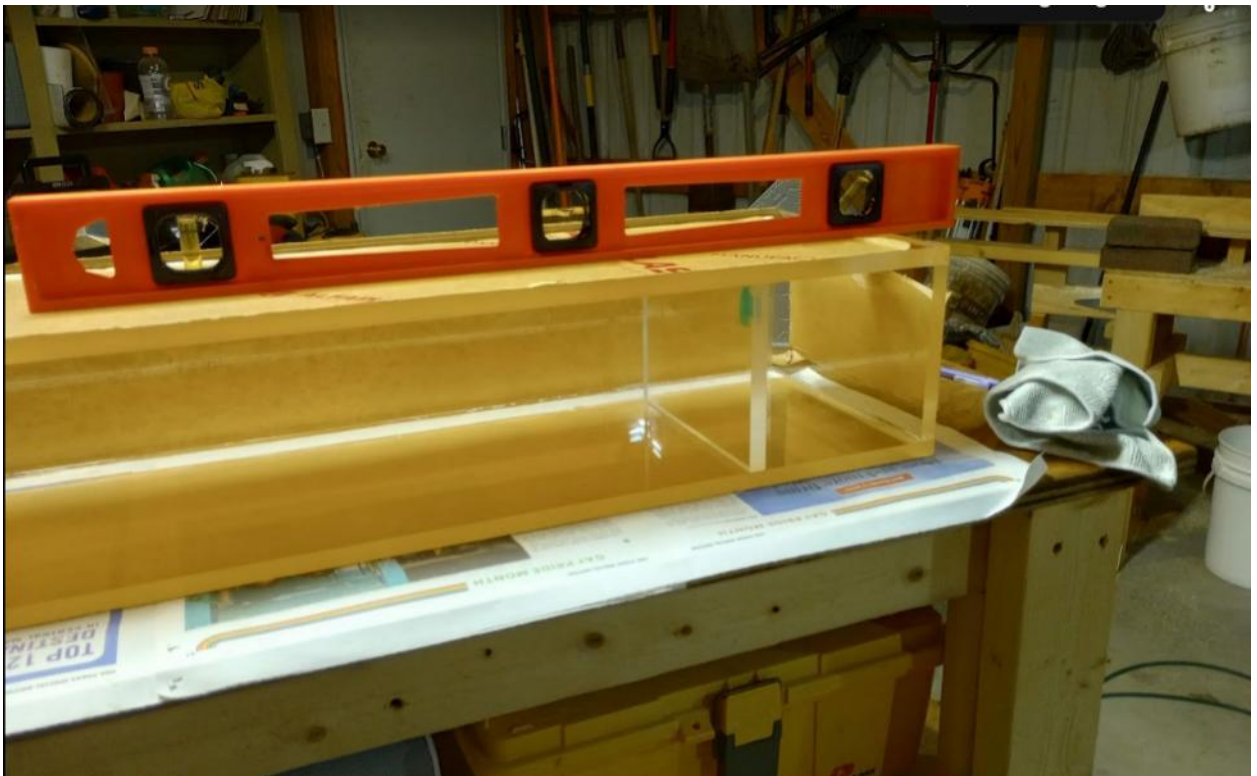


Photo 6: Ready to weld one side of the top baffle in place. Flume was laid on its side to weld inside elements.

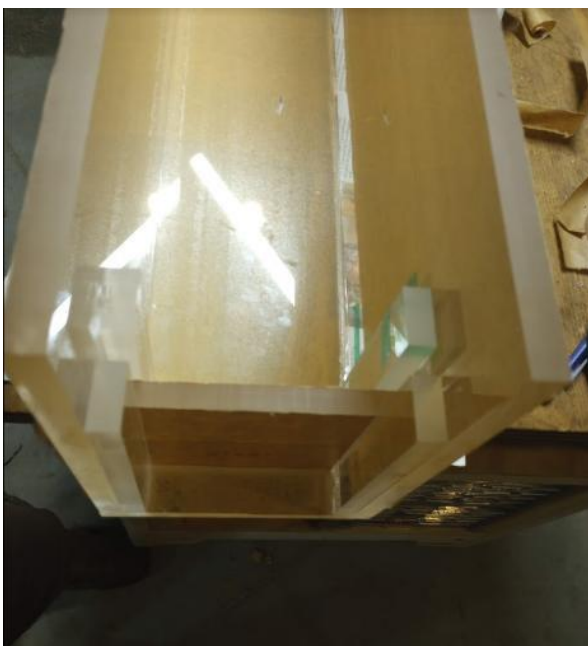


Photo 7: Preparing to weld stoplog guides in place. Paper was left on removable stoplogs to allow room for sliding them in and out. Guides were welded on one at a time. Tape was used to hold guides in place. Stoplogs were removed prior to adding solvent.



Photo 8: Preparing to weld stoplog guides in place. Guides were welded on one at a time. Tape was used to hold them in place. Stoplogs were removed prior to adding solvent.



Photo 9: Cut $\frac{1}{4}$ " acrylic ready to assemble into the weir.

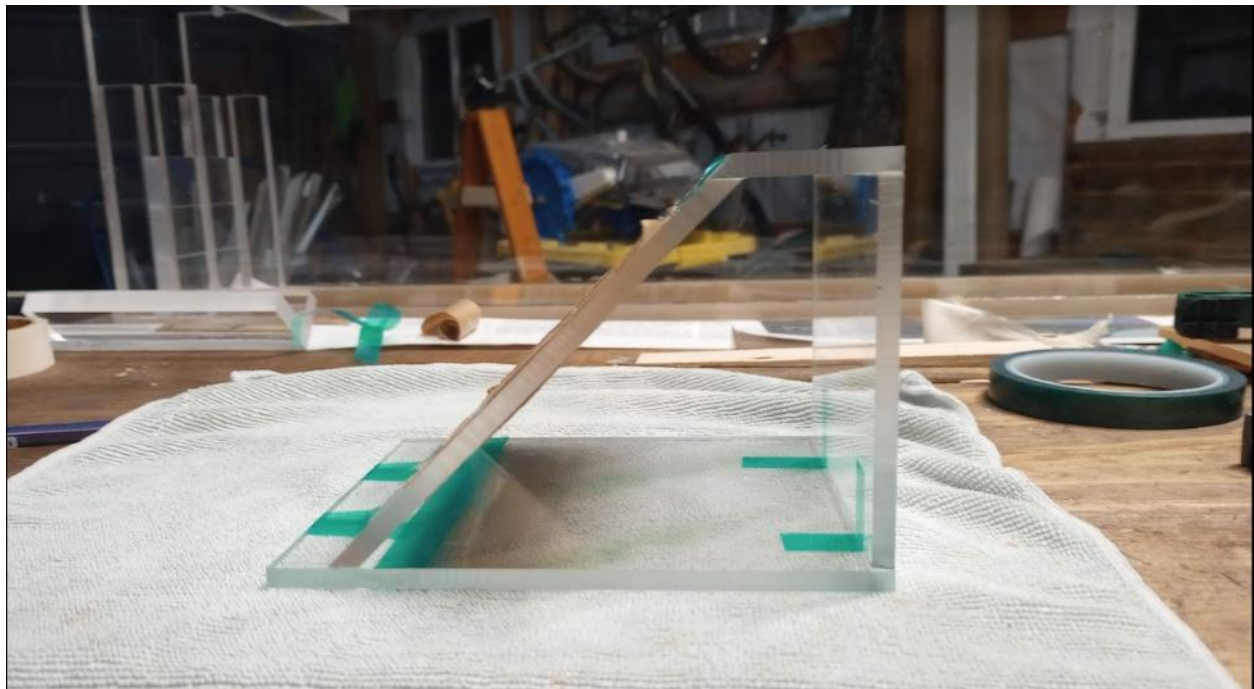


Photo 10: Weir was assembled by welding each piece on, one at a time.

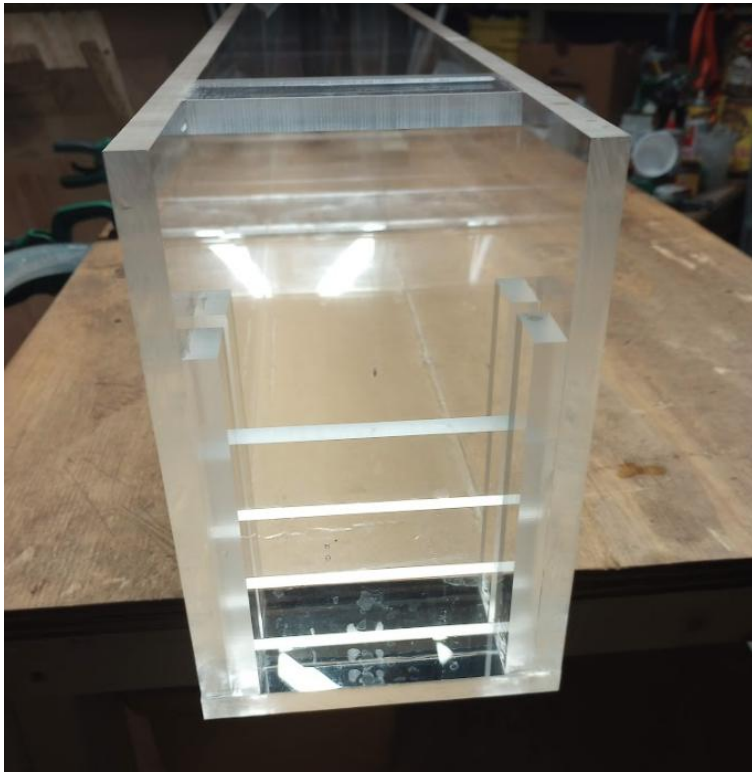


Photo 11: End view with removable stoplogs in place.

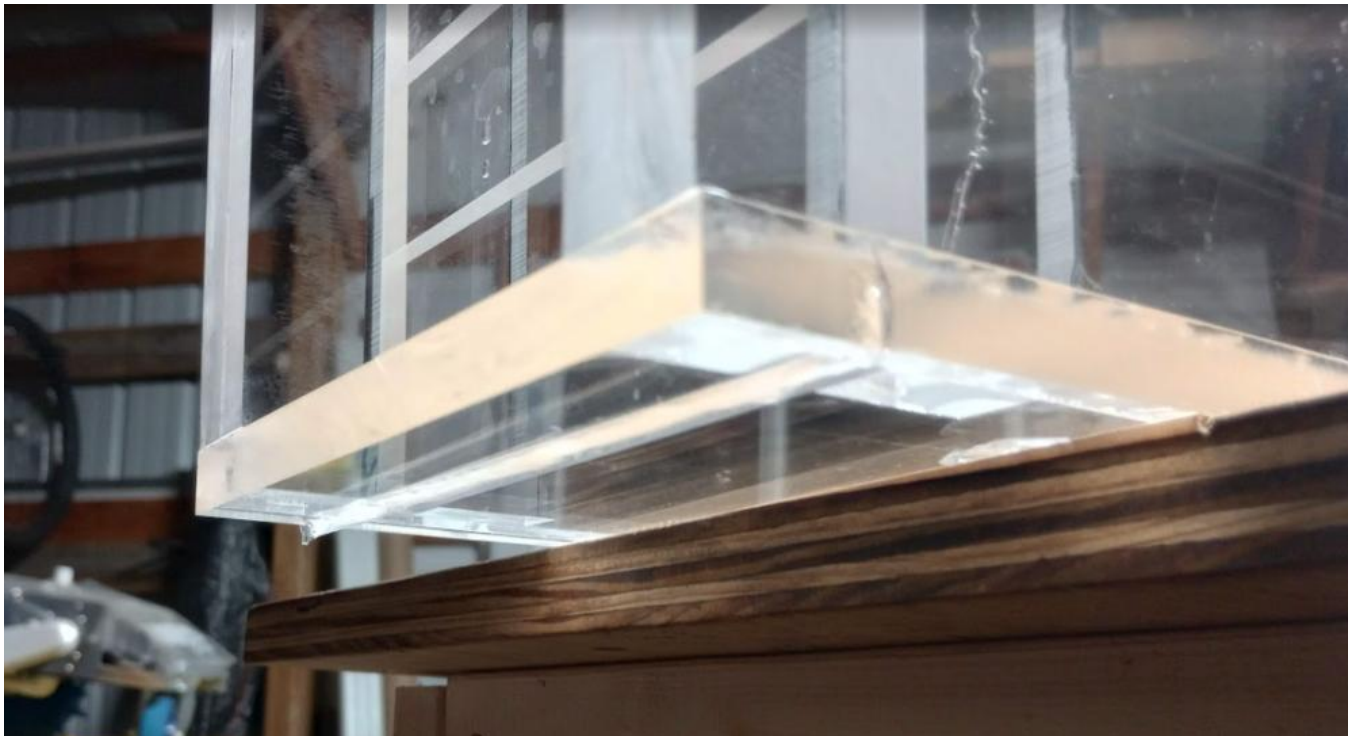


Photo 12: Silicone drip edge added to the bottom of flume on the outlet end.

Photos of Flume Setup with Pump and Hose



Photo 13: Pump with hose connection.

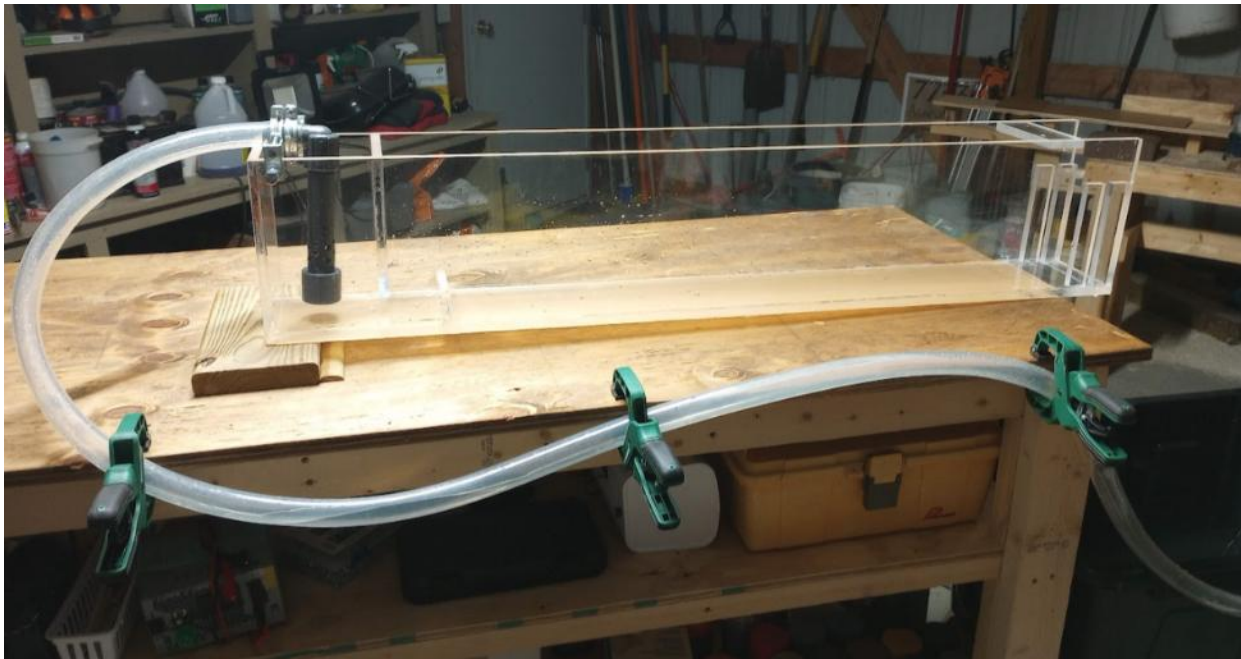


Photo 14: Hose running to the flume.



Photo 15: Mounting for hose to the flume.

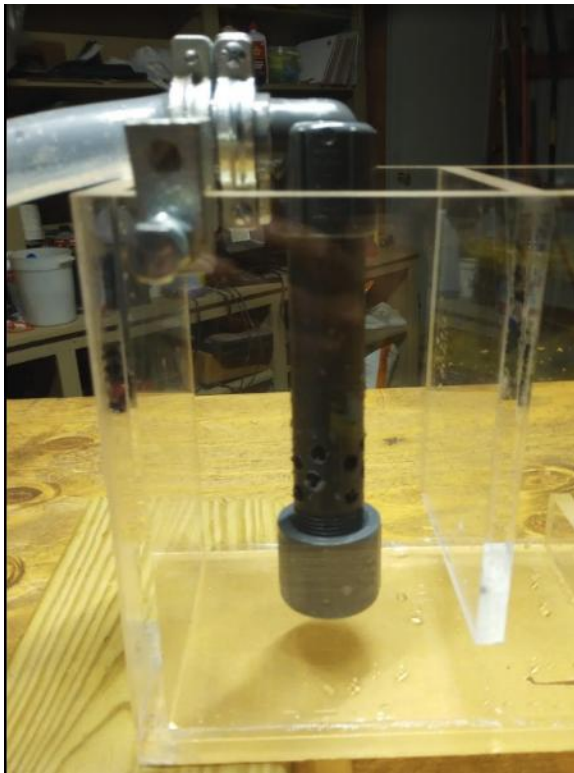


Photo 16: Side view of hose mounts and water diffuser.



Photo 17: Close up of water diffuser. Sixteen, $7/16$ " evenly spaced holes were drilled in the bottom quarter of the PVC riser. The diffuser helps distribute the incoming water evenly across the width of the flume and reduces uneven flow downstream of the inlet.